Diffusion-Augmented Deep Learning for

Colorectal Cancer Histology Classification

A Benchmark Study of CNN, Vision Transformer, and Hybrid Architectures with Stable Diffusion Minority Class Augmentation on NCT-CRC-HE

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Trial Calibrated Synthetic Data / Image Analysis

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Abstract

We present a comprehensive benchmarking study of three deep learning architectures for 9-class colorectal cancer tissue classification on the NCT-CRC-HE dataset (7,180 H&E-stained patches, 50 patients). ResNet-50, Vision Transformer (ViT-B/16), and ConvNeXt-Base are evaluated under two conditions: baseline training on real data only, and ablation training with Stable Diffusion img2img synthetic augmentation for minority tissue classes (DEB, STR). We find that diffusion augmentation improves STR classification accuracy for ResNet-50 by +3.2 percentage points but provides no benefit for ViT-B/16 and actively degrades ConvNeXt-Base performance. These architecture-dependent effects reveal that the benefit of generative augmentation depends critically on whether the architecture can already resolve the target class through its inductive bias. A FastAPI + Streamlit production deployment with GradCAM explainability, Monte Carlo Dropout uncertainty quantification, and ViT attention map visualisation is provided alongside all training artifacts and MLflow experiment tracking.

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1. Introduction and Motivation

Colorectal cancer (CRC) is the third most common cancer globally and the second leading cause of cancer-related mortality. Histopathological examination of tissue samples remains the gold standard for diagnosis, staging, and treatment planning. However, manual examination by pathologists is time-consuming, subject to inter-observer variability, and increasingly constrained by global pathologist shortages.

Computational pathology offers a compelling solution: automated classification of tissue patches extracted from hematoxylin and eosin (H&E) stained whole slide images (WSIs). Deep learning models trained on patch-level tissue classification can serve as decision-support tools, enabling faster and more consistent tissue analysis.

This project addresses two central research questions. First, which deep learning architecture best handles the 9-class tissue classification task on a small, clinically realistic dataset? Second, does Stable Diffusion-based synthetic augmentation improve classification accuracy for underrepresented tissue classes, and is this benefit architecture-dependent?

The study is motivated by the data scarcity problem inherent to medical imaging. With only 7,180 labelled patches, the dataset represents a common real-world scenario where labelled data is expensive to obtain. We investigate whether generative augmentation can compensate for this scarcity, with particular focus on the two clinically important minority classes: cancer-associated stroma (STR) and cellular debris/necrosis (DEB).

2. Dataset: NCT-CRC-HE

2.1 Overview

The NCT-CRC-HE dataset comprises 7,180 image patches extracted from H&E-stained formalin-fixed paraffin-embedded (FFPE) colorectal cancer tissue slides from 50 patients treated at the National Center for Tumor Diseases (NCT) Heidelberg and the University Medical Center Mannheim. Each patch measures 224 x 224 pixels at 0.5 microns per pixel (MPP), corresponding to 20x magnification on a standard slide scanner.

This validation set has no patient overlap with the NCT-CRC-HE-100K training set, making it a genuine out-of-distribution evaluation benchmark. The slides were Macenko color-normalized prior to tiling to reduce inter-laboratory staining variability.

2.2 Tissue Classes

The dataset contains 9 tissue classes representing distinct histological structures relevant to colorectal cancer diagnosis:

Class	Full Name	Clinical Significance	Train Samples

TUM	Colorectal adenocarcinoma epithelium	The cancer itself — malignant glandular epithelium	863
STR	Cancer-associated stroma	Reactive fibrous tissue surrounding tumor; STR/TUM ratio predicts prognosis	**295 MINORITY**
LYM	Lymphocytes	Immune response to cancer; high LYM infiltration is a positive prognostic sign	444
MUS	Smooth muscle	Colon wall muscle; TUM adjacent to MUS indicates deep invasion (worse stage)	414
NORM	Normal colon mucosa	Healthy tissue with regular gland architecture; presence indicates non-invaded regions	519
ADI	Adipose tissue	Pericolorectal fat; TUM adjacent to ADI suggests T3 staging	936
MUC	Mucus	Secreted by colon cells; abundant in mucinous adenocarcinoma subtype	724
DEB	Debris / necrosis	Dead cells and cellular fragments; common in aggressive fast-growing tumors	**237 MINORITY**
BACK	Background	Empty slide area; no biological content	593

2.3 Class Imbalance

The dataset exhibits a natural class imbalance ratio of 3.95x (ADI at 936 samples versus DEB at 237). This imbalance reflects true biological proportions: a typical CRC slide contains predominantly tumor and adipose tissue, with comparatively small regions of debris and stroma. The minority threshold was set at 75% of the median class count (556 samples), flagging DEB and STR as augmentation targets.

2.4 H&E Staining and Normalization

Hematoxylin stains cell nuclei (DNA) dark purple/blue, while eosin stains cytoplasm and connective tissue pink. The resulting dataset has a characteristic pinkish-purple color distribution, with computed dataset-specific normalization statistics of Mean [0.7268, 0.5353, 0.7092] and Std [0.1821, 0.2426, 0.1773] per channel -- substantially different from the standard ImageNet statistics ([0.485, 0.456, 0.406] / [0.229, 0.224, 0.225]) typically used for natural image models. These H&E-specific statistics were computed using Welford's online algorithm across a random subset of 7,180 patches.

Macenko normalization was applied to all slides prior to patch extraction to reduce staining variability across patients and scanners. This preprocessing step is particularly important for model generalization, as staining protocols differ substantially between laboratories.

2.5 Hard Classification Pairs

Three class pairs present particular challenges that reflect genuine diagnostic difficulty:

- STR vs MUS: Both are fibrous pink tissues. The diagnostic distinction is cell shape -- smooth muscle cells have parallel cigar-shaped nuclei, while cancer-associated stroma contains irregular spindle cells. This boundary is ambiguous even to pathologists at 20x magnification.
- LYM vs DEB: Both appear as small dark fragments at low magnification. Lymphocytes have intact round nuclei; debris lacks clear membrane integrity. The model must learn a subtle texture-level distinction.
- NORM vs well-differentiated TUM: Early-stage tumors form glands that closely resemble normal mucosa. Only mildly enlarged nuclei, small nucleoli, and subtle irregularity distinguish them.

3. Engineering Setup and Environment

3.1 Hardware

All experiments were conducted on an Apple M4 MacBook Pro using the Metal Performance Shaders (MPS) backend for PyTorch. Key MPS-specific engineering decisions included: setting pin_memory=False in all DataLoaders (Metal does not support pinned memory), using num_workers=0 to avoid multiprocessing fork issues, using torch.float16 for Stable Diffusion inference, and calling torch.mps.empty_cache() between generation steps to prevent memory pressure.

3.2 Software Stack

Component	Library	Version
Deep learning	PyTorch + torchvision	>= 2.1.0
Dataset	Hugging Face datasets	>= 2.14.0
Generative model	diffusers (Stable Diffusion)	>= 0.24.0
CLIP filtering	open-clip-torch	>= 2.23.0
Experiment tracking	MLflow	>= 2.8.0
API server	FastAPI + uvicorn	>= 0.104.0
Web interface	Streamlit	>= 1.28.0
Containerisation	Docker + docker-compose	latest

3.3 Project Structure

The project is organised as a Python package rooted at Image Analysis/ with the following module hierarchy:

- data/ -- Dataset loading, stratified splits, DataLoader construction, class statistics
- augmentation/ -- Stable Diffusion img2img pipeline, CLIP quality filter
- models/ -- ResNet-50, ViT-B/16, ConvNeXt-Base with two-stage fine-tuning utilities
- app/ -- FastAPI inference server (api.py) and Streamlit web interface (ui.py)
- train.py -- MLflow-instrumented two-stage training loop with ablation flags
- evaluate.py -- Full evaluation suite: confusion matrices, GradCAM, FID, comparison table

4. Data Pipeline

4.1 Dataset Loading

The dataset is loaded from the owkin/nct-crc-he repository on Hugging Face using split='crc_val_he_7k' to download only the 7K validation subset (~950MB). This explicit split specification was critical: the repository also contains the 100K training set, and omitting the split argument triggers a full ~10GB download. The dataset is cached locally after first download.

4.2 Stratified Splits

A 70/15/15 stratified split is applied using scikit-learn's train_test_split with stratify=labels, ensuring class proportions are preserved across train, validation, and test sets. Patient-level isolation is maintained by design -- the 7K set contains no patient overlap with the 100K training set, providing a genuine out-of-distribution test evaluation.

Final split sizes: 5,025 train / 1,077 validation / 1,078 test.

4.3 Data Augmentation

Augmentation is applied exclusively to the training split to ensure fair benchmarking across all three architectures. The augmentation pipeline includes RandomHorizontalFlip(), RandomVerticalFlip() (valid for histology -- tissue has no canonical orientation), RandomRotation(degrees=90), and ColorJitter(brightness=0.2, contrast=0.2, saturation=0.1, hue=0.05). The subtle hue jitter accounts for residual staining variation after Macenko normalization.

4.4 Normalization

All images are normalized using the dataset-specific H&E statistics computed by class_stats.py. Using ImageNet statistics would introduce a systematic bias, as the natural photo distribution (dominated by outdoor scenes) differs substantially from the pinkish-purple H&E domain.

5. Diffusion-Based Synthetic Augmentation

5.1 Motivation

Standard approaches to class imbalance include weighted sampling, class-weighted loss, SMOTE, and traditional augmentation. We instead investigate Stable Diffusion img2img augmentation, which can generate semantically diverse variations of existing patches while preserving tissue morphology. This approach is motivated by the hypothesis that generative diversity -- variations in staining intensity, cell arrangement, and tissue density -- better addresses the distribution of unseen test patches than geometric transformations alone.

5.2 Generation Pipeline

The augmentation pipeline uses Stable Diffusion v1.5 img2img (runwayml/stable-diffusion-v1-5) with the following steps:

- Source sampling: Real minority class patches are randomly sampled as init images and resized to 512x512 (SD native resolution).
- Guided generation: Each patch is processed with a class-specific H&E pathology prompt and a strength parameter controlling the degree of deviation from the source image (DEB: 0.35, STR: 0.40). Lower strength preserves tissue structure while introducing plausible variation.
- Quality filtering: Each generated image is scored by CLIP cosine similarity against the class prompt using OpenCLIP ViT-B/32. Images scoring below 0.22 are discarded.
- Storage: Accepted images are saved to outputs/synthetic/{CLASS}/ as PNG files for reuse across training runs.

5.3 Class-Specific Prompts

Prompts were carefully engineered to stay within the H&E histology domain:

- DEB: 'hematoxylin and eosin stained histology, debris and necrotic tissue, colorectal cancer, cellular fragments, eosinophilic material, 224x224 microscopy patch, high quality pathology slide'
- STR: 'hematoxylin and eosin stained histology, cancer-associated stroma, colorectal adenocarcinoma, spindle cells, fibrous connective tissue, desmoplastic stroma, 224x224 microscopy patch, high quality pathology slide'

A universal negative prompt steered generation away from artefacts: 'blurry, low quality, oversaturated, cartoon, drawing, text, out of focus, jpeg artefacts, watermark'.

5.4 Generation Results

100 synthetic patches were generated per minority class (200 total), with a 100% CLIP acceptance rate at threshold 0.22. Generation time was approximately 10 seconds per image on Apple M4 MPS (35 minutes total). FID scores computed using InceptionV3 features were 423.59 for DEB and 466.72 for STR, indicating a measurable domain gap relative to real

images. For context, well-calibrated generative models on natural images achieve FID < 10; for histology the acceptable threshold is higher due to CLIP's underrepresentation of medical imaging in its training data.

5.5 Biological Considerations

DEB patches carry higher biological synthesis risk than STR patches. Real debris results from specific biological processes (necrosis, apoptosis) with characteristic nuclear fragmentation patterns that Stable Diffusion approximates through texture matching rather than biological fidelity. STR, by contrast, has more regular fibrous structure that generative models can plausibly reproduce. The CLIP filter mitigates but does not eliminate this risk -- CLIP operates on semantic similarity, not biological accuracy.

6. Model Architectures

6.1 ResNet-50 (CNN Baseline)

ResNet-50 serves as the convolutional baseline. Its design encodes a strong spatial inductive bias through sliding 3x3 filters that assume locality (nearby pixels are related) and translation invariance (patterns mean the same thing anywhere in the image). These assumptions are well-suited to histology, where tissue structures are spatially coherent.

The custom classification head replaces the original 1000-class fully connected layer with: Dropout(0.3) -> Linear(2048, 512) -> ReLU -> Dropout(0.3) -> Linear(512, 9). Total parameters: ~25M. Pretrained weights: ImageNet1K_V2.

6.2 Vision Transformer ViT-B/16

ViT-B/16 divides the 224x224 image into 196 non-overlapping 16x16 patches, embeds each as a 768-dimensional vector, prepends a learnable [CLS] token, and processes the resulting 197-token sequence through 12 transformer encoder layers with 12 attention heads each. Self-attention allows any patch to directly attend to any other patch regardless of spatial distance.

This architecture has no spatial inductive bias -- it must learn from data that nearby patches are related. Consequently, ImageNet features transfer less directly (stage 1 accuracy: 89.3% vs ResNet's 97.9%), but the global attention mechanism provides a meaningful advantage for tissue classes whose diagnostic signal is distributed across the whole patch rather than localised to specific regions.

The classification head: Dropout(0.1) -> Linear(768, 256) -> GELU -> Dropout(0.1) -> Linear(256, 9). Total parameters: ~86M. Pretrained weights: ImageNet1K_SWAG_LINEAR_V1.

6.3 ConvNeXt-Base (Hybrid)

ConvNeXt-Base is described as 'hybrid' because it retains the CNN structural hierarchy (four stages of progressively larger receptive fields) but replaces standard convolutions with transformer-inspired design choices: depthwise separable convolutions, inverted bottlenecks, LayerNorm instead of BatchNorm, and GELU activations.

The effect is a model that combines the data efficiency and spatial inductive bias of CNNs with the representational capacity of transformers. It consistently outperforms both pure CNNs and ViTs on small-to-medium pathology datasets in published benchmarks. Classification head: Dropout(0.2) -> Linear(1024, 256) -> GELU -> Dropout(0.2) -> Linear(256, 9). Total parameters: ~89M.

7. Training Methodology

7.1 Two-Stage Fine-Tuning

All models are trained using a two-stage strategy motivated by the small dataset size (5,025 training samples) and the catastrophic forgetting risk associated with full fine-tuning on limited data.

Stage 1 (frozen backbone): The backbone weights are frozen by setting requires_grad=False on all non-head parameters. Only the classification head (~1-1.3M parameters) is trained. This fast convergence stage leverages the pretrained features directly without modifying them, reaching high accuracy within 5 epochs.

Stage 2 (partial unfreeze): The top backbone layers are unfrozen with a 10x lower learning rate than the head. Differential learning rates prevent catastrophic forgetting -- backbone weights are nudged gently toward histology-specific features while the head continues to adapt. Layer selection varies by architecture: layer4 for ResNet-50 (15M parameters), last 4 of 12 transformer blocks for ViT-B/16 (28.5M parameters), and last feature stage for ConvNeXt-Base (27.7M parameters).

7.2 Training Configuration

Parameter	ResNet-50	ViT-B/16	ConvNeXt
Stage 1 epochs	5	5	5
Stage 2 epochs	10	15	10
Stage 1 LR	1e-3	3e-5	5e-4
Stage 2 LR	1e-4	1e-5	5e-5
Batch size	32	16	32
Optimizer	AdamW	AdamW	AdamW
Scheduler	CosineAnnealingLR	CosineAnnealingLR	CosineAnnealingLR
Loss function	CrossEntropy (label smoothing 0.1)	CrossEntropy (label smoothing 0.1)	CrossEntropy (label smoothing 0.1)

ViT requires both more epochs and a lower learning rate than ResNet and ConvNeXt. Transformers are sensitive to learning rate -- too high a rate causes training instability in the attention layers. The smaller batch size for ViT reflects its higher memory footprint on MPS.

7.3 MLflow Experiment Tracking

All hyperparameters, per-epoch metrics, and final test results are logged to MLflow under the crc-histology-benchmark experiment. Each run is named with the convention {model}_{condition}_seed42. Model artifacts are registered in the MLflow model registry as crc_cnn, crc_vit, and crc_hybrid, each with two versions (baseline and synthetic). The MLflow UI provides training curve visualisation, parallel coordinate plots for hyperparameter comparison, and artifact download.

8. Evaluation Framework

8.1 Metrics

Primary metric: per-class accuracy on the held-out test set (1,078 patches). This is more informative than overall accuracy for imbalanced datasets -- a model achieving 99% overall accuracy by ignoring the 51-sample DEB class would appear competitive but be clinically useless.

Secondary metrics: macro-averaged F1 score (equal weight to all classes regardless of support), classification report (precision, recall, F1 per class), and normalised confusion matrix. The STR delta metric captures the change in STR accuracy between baseline and synthetic conditions, quantifying the ablation effect directly.

8.2 GradCAM Explainability

Gradient-weighted Class Activation Mapping (GradCAM) is computed for CNN and ConvNeXt models by registering forward and backward hooks on the last convolutional feature map. The gradient of the predicted class score with respect to the feature map is globally average-pooled to produce channel weights, which are multiplied by the feature map activations and ReLU-activated to produce a spatial heatmap. This heatmap is resized to 224x224 and overlaid on the original patch, showing which spatial regions most influenced the prediction.

8.3 Monte Carlo Dropout Uncertainty

Epistemic uncertainty is estimated using Monte Carlo Dropout. The model is set to training mode (activating dropout stochasticity) and 20 stochastic forward passes are run per image. The standard deviation across passes serves as an uncertainty proxy. A flag_for_review signal is raised when max std exceeds 0.08, indicating the model encountered an input unlike its training distribution. This is a clinically critical feature -- a classifier that reports high confidence on all inputs is unsafe for deployment.

8.4 ViT Attention Analysis

ViT attention weights are extracted by monkey-patching each encoder layer's nn.MultiheadAttention.forward to force need_weights=True, average_attn_weights=False. This returns the full [batch, n_heads, seq_len, seq_len] attention weight tensor per layer. The CLS token's attention to all patch tokens from the last encoder layer is used to produce an attention rollout map (reshaped to 14x14). Per-head entropy is computed to characterise head specialisation: low entropy indicates a focused head attending to specific tissue regions, high entropy indicates diffuse global attention.

8.5 FID Score

The Frechet Inception Distance between real and synthetic image distributions is computed using InceptionV3 feature embeddings. FID measures the distance between the mean and covariance of the two feature distributions. Lower FID indicates more realistic synthetic images. Note that our implementation uses a simplified FID (diagonal covariance approximation) for computational efficiency -- the full matrix square root is expensive for small sample sizes.

9. Results and Analysis

9.1 Baseline Performance

Model	Test Accuracy	Macro F1	STR Accuracy	DEB Accuracy
ResNet-50	98.6%	98.5%	92.1%	98.0%
ViT-B/16	98.8%	98.5%	**95.2%**	98.0%
ConvNeXt	98.6%	98.4%	92.1%	98.0%

9.2 Key Baseline Observations

Stage 1 accuracy divergence is the most informative early signal. ResNet-50 achieved 97.9% validation accuracy after 5 frozen epochs, while ViT-B/16 reached only 89.3%. This gap directly reflects the spatial inductive bias advantage: ResNet's pretrained sliding filters immediately produce useful H&E features, while ViT must re-learn spatial relationships from limited data before its global attention becomes effective.

DEB accuracy is identical across all three architectures at 98.0% (1 misclassified patch out of 51). This suggests DEB is primarily a local texture classification problem that all architectures solve equally well. The diagnostic signal -- fragmented membranes, karyolytic nuclei -- is captured by local filters as effectively as global attention.

STR is the discriminating class. ViT outperforms both ResNet and ConvNeXt by 3.1 percentage points (95.2% vs 92.1%). Cancer-associated stroma is diagnosed by the overall tissue arrangement -- disordered fibrous patterns distributed across the whole patch -- rather than any localised feature. ViT's global self-attention is architecturally better suited to integrate this distributed signal.

ConvNeXt matching ResNet on STR despite its hybrid design is informative. Although ConvNeXt incorporates transformer-style design principles, it retains the CNN structural hierarchy and local receptive fields. The global context ViT achieves through self-attention is not replicated by depthwise convolutions alone, even with a transformer-like architecture.

10. Ablation Study: Effect of Synthetic Augmentation

10.1 Full Results

Model	Condition	Test Acc	STR Acc	DEB Acc	STR Delta
ResNet-50	Baseline	98.6%	92.1%	98.0%	--
ResNet-50	**+Synthetic**	98.9%	**95.2%**	98.0%	**+3.1%**
ViT-B/16	Baseline	98.8%	95.2%	98.0%	--
ViT-B/16	**+Synthetic**	98.9%	95.2%	98.0%	0.0%
ConvNeXt	Baseline	98.6%	92.1%	98.0%	--
ConvNeXt	**+Synthetic**	98.1%	**90.5%**	94.1%	**-1.6%**

10.2 ResNet-50: Synthetic Augmentation Helps (+3.1% STR)

Adding 100 synthetic STR patches lifted ResNet-50's STR accuracy from 92.1% to 95.2%, matching ViT's baseline performance. Overall accuracy also improved marginally (98.6% to 98.9%). This confirms the central hypothesis: CNN architectures, which rely on local filters and cannot natively integrate global tissue context, benefit from additional training examples of the globally-distributed STR pattern.

Mechanistically, the synthetic patches gave the deeper layers of ResNet (layer4) more examples of the disordered fibrous pattern characteristic of cancer-associated stroma. After fine-tuning, the feature representations for STR became more discriminative, reducing the confusion with MUS (the primary confounding class).

10.3 ViT-B/16: No Effect (0.0% STR Delta)

ViT's STR accuracy was unchanged by synthetic augmentation. This negative result is theoretically significant: ViT's global self-attention already captures the tissue-wide arrangement that defines STR, achieving ceiling performance on this class without additional data. The synthetic patches added no new information that the architecture could not already extract from the original 295 STR training samples.

This finding suggests that for classes defined by global tissue architecture (as opposed to local cellular features), architectural choice is a more powerful lever than data augmentation.

10.4 ConvNeXt: Synthetic Augmentation Hurts (-1.6% STR, -3.9% DEB)

ConvNeXt is the most architecturally sensitive to the domain gap introduced by synthetic patches. Both STR and DEB accuracy decreased, with the DEB drop (98.0% to 94.1%) being particularly notable. ConvNeXt's hybrid design -- transformer-style normalization and attention within a CNN hierarchy -- makes it more sensitive to subtle distributional shifts in training data compared to pure ResNet.

The FID scores (DEB: 423.59, STR: 466.72) confirm a measurable domain gap between real and synthetic images. ResNet's local convolutional filters are relatively insensitive to this global statistical difference. ConvNeXt's layer normalization and depthwise convolutions, which operate on channel statistics, are more affected. The synthetic patches effectively introduced a confounding distribution that degraded the model's calibration for both minority classes.

This result carries an important practical implication: diffusion augmentation is not universally beneficial and should be validated per-architecture before deployment. Blindly applying generative augmentation to a ConvNeXt-style architecture on this dataset would have degraded performance.

11. Explainability and Uncertainty Quantification

11.1 GradCAM Analysis

GradCAM heatmaps were generated for 3 DEB and 3 STR patches per model across all 6 training conditions (36 total images saved to outputs/evaluation/gradcam/). CNN and ConvNeXt heatmaps show activation concentrated on cellular structures -- nuclear clusters for TUM predictions, fibrous regions for STR predictions. The heatmaps provide a visual sanity check that the model is attending to biologically relevant features rather than artifacts.

11.2 ViT Attention Rollout

The ViT attention map shows the [CLS] token's aggregate attention to all 196 patch tokens from the last encoder layer. For TUM patches, attention is broadly distributed across the patch, reflecting the diagnostic reliance on overall glandular architecture rather than localised features. Head entropy values of 4.1-4.7 across all 12 heads confirm diffuse attention for this tissue class.

The monkey-patching approach used to extract attention weights (overriding nn.MultiheadAttention.forward to force need_weights=True, average_attn_weights=False) was necessary because torchvision's ViT implementation discards attention weights during standard inference. This is a known limitation of the torchvision implementation; the HuggingFace transformers library provides a cleaner output_attentions=True interface.

11.3 Monte Carlo Dropout Uncertainty

20-pass MC Dropout uncertainty estimation is integrated into the inference pipeline. The flag_for_review threshold of 0.08 (max std across classes) was set empirically to flag approximately the most uncertain 5% of predictions. In clinical deployment, flagged patches would be routed to a pathologist for manual review, creating a human-in-the-loop system rather than a fully autonomous classifier.

The uncertainty panel in the Streamlit interface displays per-class std alongside mean confidence, allowing users to see not just which class was predicted but how stable that prediction is across stochastic forward passes. A prediction of TUM at 94% mean confidence with std 0.01 is clinically very different from the same prediction with std 0.12.

12. Production Deployment

12.1 FastAPI Inference Server

The inference server (app/api.py) exposes four endpoints:

- GET /health -- liveness check returning device type and number of loaded models
- POST /predict -- standard inference returning predicted class, confidence, all class probabilities, and inference time
- POST /predict/gradcam -- inference with GradCAM heatmap returned as base64-encoded PNG
- POST /predict/uncertain -- Monte Carlo Dropout inference returning mean/std per class and flag_for_review signal
- POST /predict/attention -- ViT-only endpoint returning attention rollout heatmap and 12 per-head entropy values

Models are loaded on first request and cached in memory. Differential learning rates and two-stage training are abstracted away from the API -- callers specify model name and condition (baseline/synthetic) and receive identical response schemas regardless of the underlying architecture.

12.2 Streamlit Web Interface

The web interface (app/ui.py) provides a clinical-facing demonstration with four panels: original patch display, GradCAM heatmap overlay, prediction card with confidence and inference time, and a probability bar chart for all 9 classes. An uncertainty panel shows MC Dropout results with the flag_for_review signal displayed as a green checkmark or amber warning. The ViT attention panel shows head entropy bars when ViT-B/16 is selected.

The sidebar displays live benchmark metrics from evaluation_report.json for the currently selected model and condition, providing immediate context for the prediction confidence.

12.3 Docker Deployment

A Dockerfile and docker-compose.yml are provided for reproducible deployment. docker-compose up starts both the FastAPI server (port 8000) and Streamlit UI (port 8501) with shared volume mounting for model checkpoints. The API container includes a health check that blocks the UI container from starting until the API reports healthy, preventing race conditions during startup.

13. Engineering Challenges and Solutions

Challenge 1: Dataset Download Size

Problem: The owkin/nct-crc-he repository bundles three splits (100-sample, 1K-sample, and 7K-sample) in a single Hugging Face repository. Without explicit split specification, the datasets library downloads all splits (~10GB). Initial attempts with the DykeF/NCT-CRC-HE-100K repository triggered a full 21GB download.

Solution: Switched to owkin/nct-crc-he with explicit split='crc_val_he_7k'. This required iterative debugging -- the 1aurent/NCT-CRC-HE repository ignored split parameters due to a custom loading script, downloading all splits regardless. Confirmed the correct split name using streaming=True before committing to a full download.

Challenge 2: MPS Batch Processing in Stable Diffusion

Problem: Passing a list of PIL images to the SD img2img pipeline raised 'Expected size for first two dimensions of batch2 tensor to be: [1, 40] but got: [0, 40]' on MPS. The error indicated that batch matrix multiplication was failing with the MPS attention processor.

Solution: Process images one at a time in a loop rather than as a batch. While slower, this is stable on all MPS configurations and produces identical outputs. Generation time increased from ~8s/image to ~10s/image -- acceptable for a one-time offline augmentation step.

Challenge 3: GradCAM Tensor Device Mismatch

Problem: GradCAM hook outputs on MPS raised 'can't convert mps:0 device type tensor to numpy'. PyTorch's MPS tensors cannot be directly converted to NumPy arrays.

Solution: Added .cpu().detach() before .numpy() on all gradient and activation tensors: grads = self.gradients[0].cpu() and F.relu(torch.as_tensor(heatmap).cpu().detach()).numpy(). This ensures tensors are moved from MPS device to host memory before NumPy conversion.

Challenge 4: ViT Attention Weight Extraction

Problem: Torchvision's ViT implementation calls nn.MultiheadAttention with need_weights=False by default, discarding attention weights during inference. Standard forward hooks capture the attention output tensor (post-projection) rather than the attention weight matrix, resulting in all heads returning entropy 1.00.

Solution: Monkey-patched each encoder layer's self_attention.forward method to force need_weights=True, average_attn_weights=False. The patched forward captures the [batch, n_heads, seq_len, seq_len] attention weight matrix before it is discarded. A closure-based approach (make_patched(orig, extractor)) ensures each layer's original forward method is correctly preserved in the closure.

Challenge 5: models Package Namespace Conflict

Problem: Python resolved 'from models import build_model' to a system-level models namespace package rather than the project's models/ directory, raising ImportError. The __init__.py file was present but the namespace resolution prioritised the conda environment.

Solution: Added sys.path.insert(0, str(Path(__file__).resolve().parent)) at the top of train.py and api.py to force the local package to take precedence. Additionally ensured models/__init__.py was correctly populated with all exports.

Challenge 6: MLflow Path Spaces

Problem: The project path contains spaces ('Trial Calibrated Synthetic Data/Image Analysis'), causing mlflow ui --backend-store-uri file://$(pwd)/outputs/mlruns to fail with 'unexpected extra arguments'.

Solution: Quoted the URI: mlflow ui --backend-store-uri "file://$(pwd)/outputs/mlruns" --port 5000.

14. Discussion

14.1 Architecture Choice vs Data Augmentation

The most important finding of this study is that architectural choice is a stronger determinant of minority class accuracy than data augmentation for this dataset. ViT-B/16 achieves 95.2% STR accuracy without any synthetic data, while ResNet-50 requires synthetic augmentation to reach the same level. ConvNeXt is harmed by augmentation that helps ResNet.

This has a practical implication for computational pathology practitioners: before investing in generative augmentation pipelines, the choice of architecture should be evaluated first. A ViT with appropriate fine-tuning may solve the minority class problem without the complexity, cost, and risk of generative augmentation.

14.2 Distribution Gap and Architecture Sensitivity

The FID scores (423-466) reveal a meaningful gap between real and synthetic image distributions. The fact that ResNet tolerates this gap while ConvNeXt does not suggests that CNN architectures with standard BatchNorm and local filters are more robust to distributional shift in training data than architectures with LayerNorm and global-channel operations. This is a non-obvious result that warrants further investigation with controlled synthetic data quality experiments.

14.3 Clinical Deployment Considerations

The uncertainty quantification component (MC Dropout) is the most important feature for clinical deployment. A model that is always confident is clinically dangerous -- it provides no signal for cases where human review is needed. The flag_for_review mechanism creates a natural human-in-the-loop workflow where high-uncertainty patches are escalated to pathologists rather than making autonomous predictions.

The GradCAM and attention visualisations serve a different purpose: building clinician trust. When a model predicts TUM and the heatmap shows activation on the irregular glandular structures that a pathologist would also identify, it validates that the model is reasoning about tissue biology rather than spurious correlations.

15. Conclusions and Future Work

15.1 Conclusions

We have demonstrated that:

- ViT-B/16's global self-attention provides a meaningful advantage over CNN-based architectures for cancer-associated stroma classification, where the diagnostic signal is distributed across the whole tissue patch.
- Stable Diffusion img2img augmentation improves CNN performance on minority classes (+3.1% STR for ResNet-50) by compensating for the CNN's inability to integrate global tissue context.
- Diffusion augmentation is architecture-dependent: it provides no benefit for ViT and actively degrades ConvNeXt, likely due to the distributional gap between synthetic and real patches interacting differently with each architecture's normalisation and feature extraction mechanisms.
- FID scores of 423-466 confirm a measurable domain gap in our synthetic images, highlighting the importance of higher-fidelity generative models (trained specifically on pathology data) for medical image augmentation.
- Monte Carlo Dropout uncertainty quantification is a practical addition to any clinical deployment, enabling human-in-the-loop review of uncertain predictions.

15.2 Future Work

Several directions would meaningfully extend this work:

- Pathology foundation models: Replacing ImageNet pretrained weights with UNI or CONCH (pretrained on 100K+ pathology slides) would likely push accuracy to 99%+ and provide more clinically meaningful feature representations.
- Higher-fidelity generative models: Training a domain-specific diffusion model on H&E histology data (rather than using general SD v1.5) would reduce the FID gap and potentially allow ConvNeXt to benefit from synthetic augmentation.
- Slide-level aggregation: Extending patch-level predictions to a whole-slide tumor map, where thousands of patch predictions are aggregated into a tissue composition heatmap and slide-level risk score.
- Self-supervised pretraining: Pretraining a ViT on the unlabelled NCT-CRC-HE-100K dataset using DINO or SimCLR before supervised fine-tuning could improve performance on the 7K labelled set.
- Attention head specialisation: A systematic analysis of which ViT attention heads specialise for which tissue structures would provide mechanistic insight into how transformers learn histological representations.

16. References

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